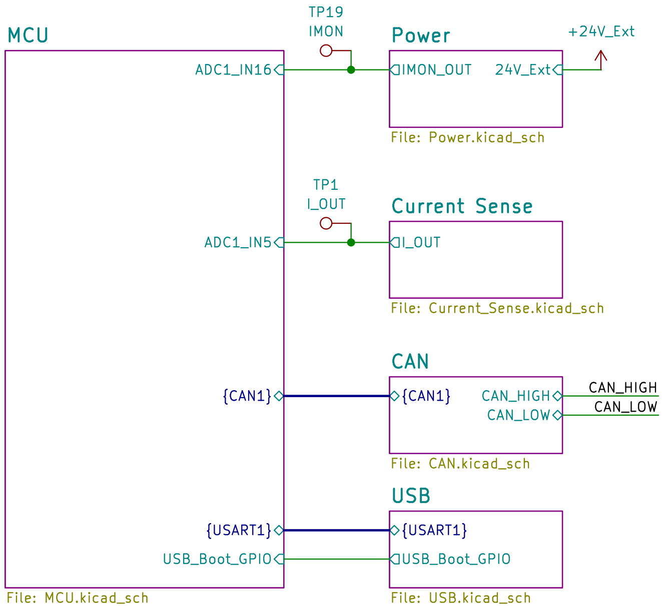

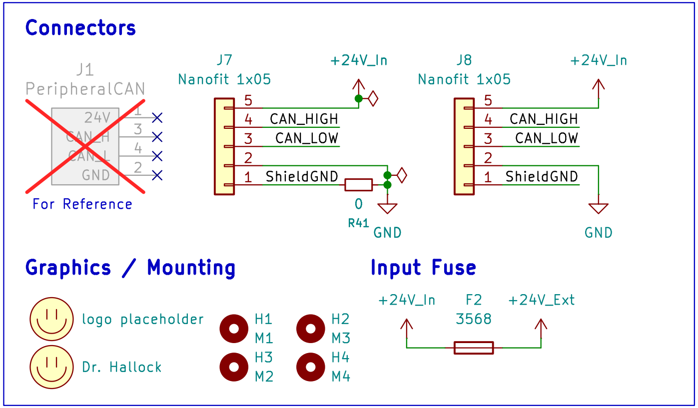

RevB Changelog

- INA: tied SHUNT−/Scrut− to GND HV (INA GND) for consistent common−mode voltage
- AMC / TLV: converted differential output to single−ended via differential amp / LPF
- Scrutineering: SPDT switch, standardized resistors
- USB: Horizontal footprint, CP2102N GPIO pin for boot
- MCU: enable bootloader using CP2102N GPIO, MCU GPIO, or switch; diode OR
- Power: Removed SHDN resistor; fuses for input and isolated DC−DC buck
- Debug: added VBUS LED, standardized LED resistors
- Changed to shielded connectors

Board Notes

Measures current from battery using low−side shunt resistor
− Note for charging: max 32A thru Elcon, max 54A to cell

I_OUT = (current * shunt * gain) + Vref (V)
= (current * 0.025) + 1.25 (V)

Battery Current Range	Current (A)	I_OUT Reading (V)
Max Charge (−)	−32 A	0.45 V
Max Discharge (+)	+70 A	3.0 V

Measurable Current Range	Current (A)	I_OUT Reading (V)
Max Charge (−)	−46 A	0.1 V
Max Discharge (+)	+78 A	3.2 V

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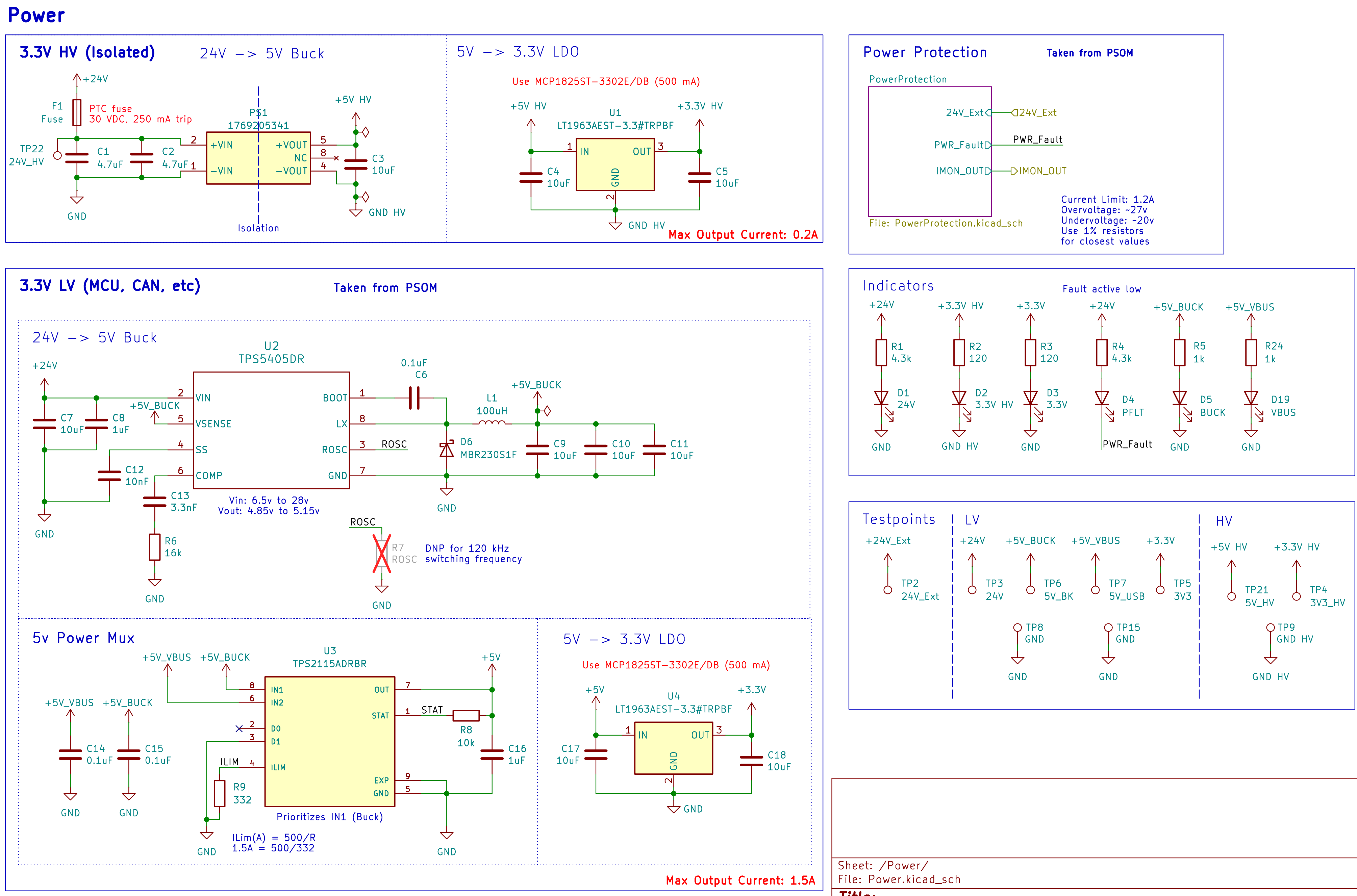

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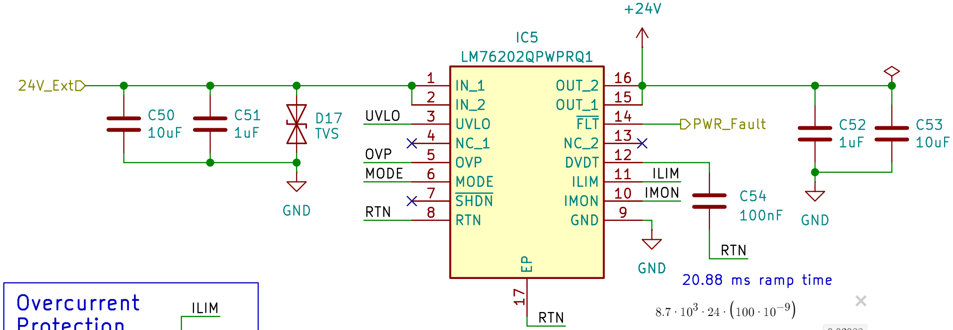
Taken from PSOM



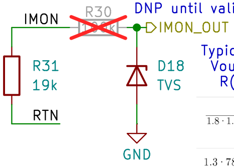
20.88 ms ramp time

$$8.7 \cdot 10^3 \cdot 24 \cdot \left(100 \cdot 10^{-9}\right) = 0.02088$$

$$t_{dVdT} = 8.7 \times 10^3 \times V_{(IN)} \times C_{(dVdT)}$$

RTN!=GND
Conecting GND to RTN turns off RPP!

Current Monitoring



Typical gain: 78.28 uA/A
Vout @ 1.3A = 2.407v
R(imon) = 23.7 mΩ

$$\frac{4}{1.8 \cdot 1.2 \cdot 78.28} = 0.0236567686747$$

$$1.3 \cdot 78.28 \cdot 0.0236567686747 = 2.40740740741$$

$$\left(2 \cdot 10^{-6}\right)\left(0.0236567686747\right) = 4.7313537349 \times 10^{-8}$$

$$R_{(IMONmax)} = \frac{\text{Min}\left[\left(V_{(IN)} - 1.5\right), 4\text{ V}\right]}{1.8 \times I_{(LIM)} \times GAIN_{(IMON)}}$$

For $I_{OUT} > 50$ mA,

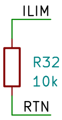
$$V_{(IMON)} = \left[I_{(OUT)} \times GAIN_{(IMON)}\right] \times R_{(IMON)}$$

For $I_{OUT} < 50$ mA (typical), IMON output current is close to $I_{(MON\_OS)}$ with R_{IMON}.

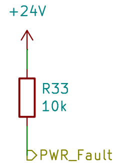
$$V_{(IMON)} = \left(I_{(IMON\_OS)}\right) \times R_{(IMON)}$$

Overcurrent Protection

ILim(A) = 12/R(k)
1.2A = 12/10kΩ

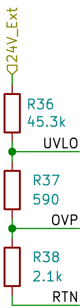


Pullups



RIP shdn resistor

Under/Overvoltage Protection



	Setpoint	Ideal	Actual
UVLO	20v	1.1v	1.12v
OVP	27v	1.19v	1.18v

OVP set to 27v for
buck max voltage of 28v

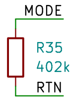
$$\left(\frac{2.1 + 0.59}{2.1 + 0.59 + 45.3}\right)20 = 1.12106688894$$

$$\left(\frac{2.1}{2.1 + 0.59 + 45.3}\right)27 = 1.18149614503$$

Functional Mode Selection

Set to circuit breaker (latchoff)

Use 0 ohm resistor for
auto-retry (540ms)



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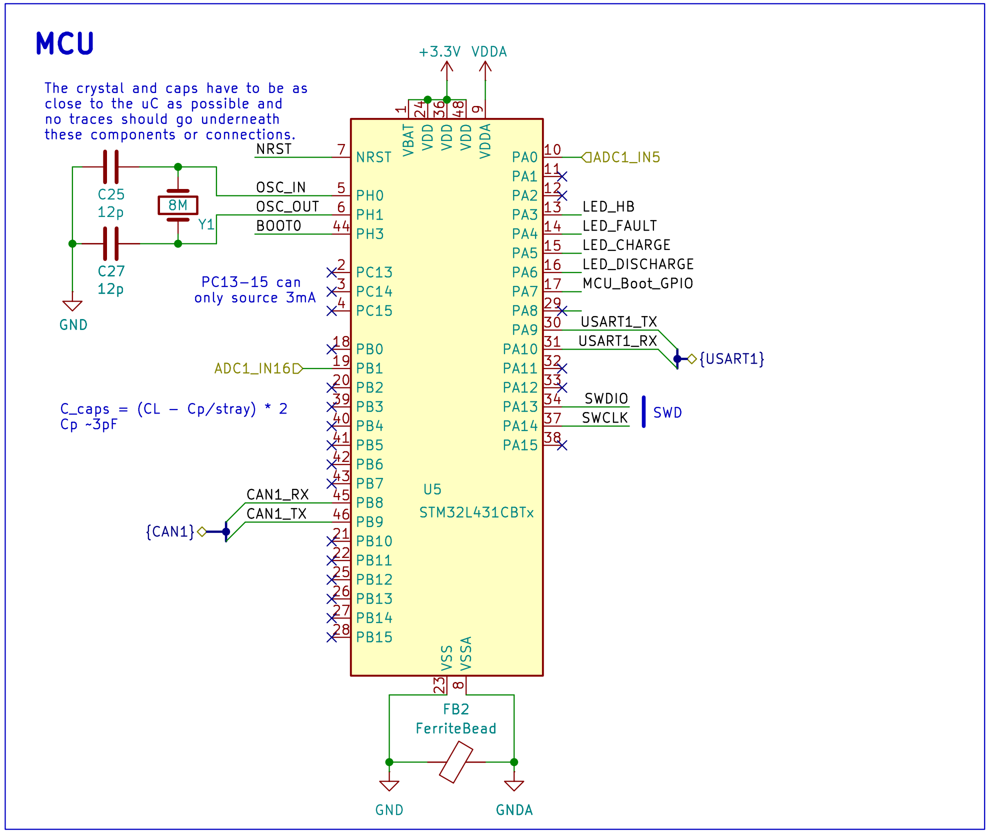
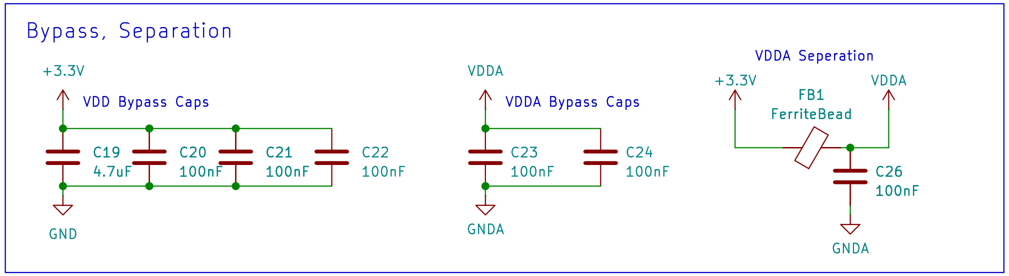
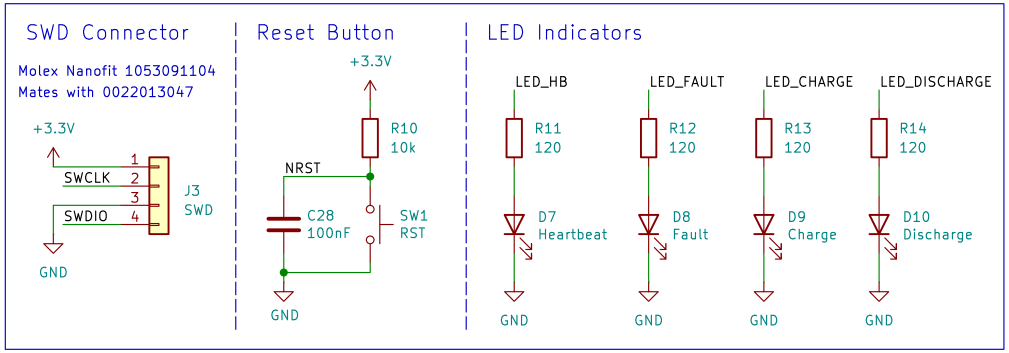
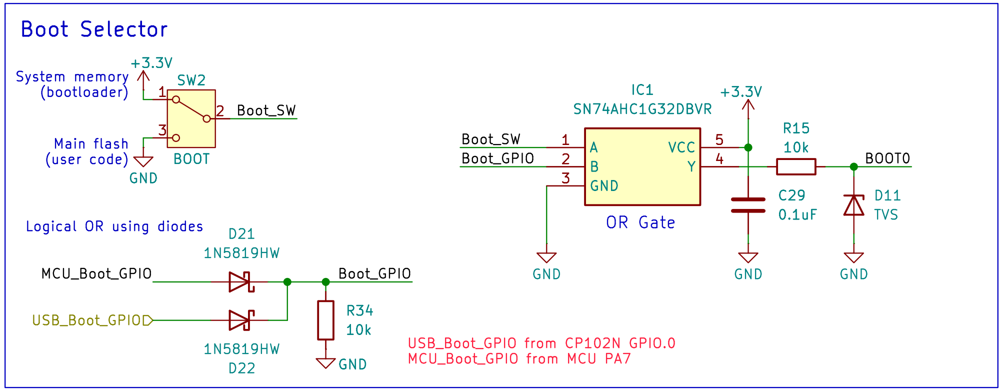


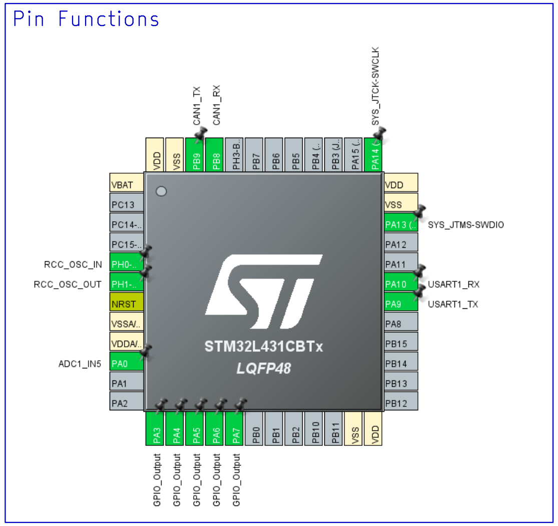




USART bootloader pins (same as PSOM)

USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.





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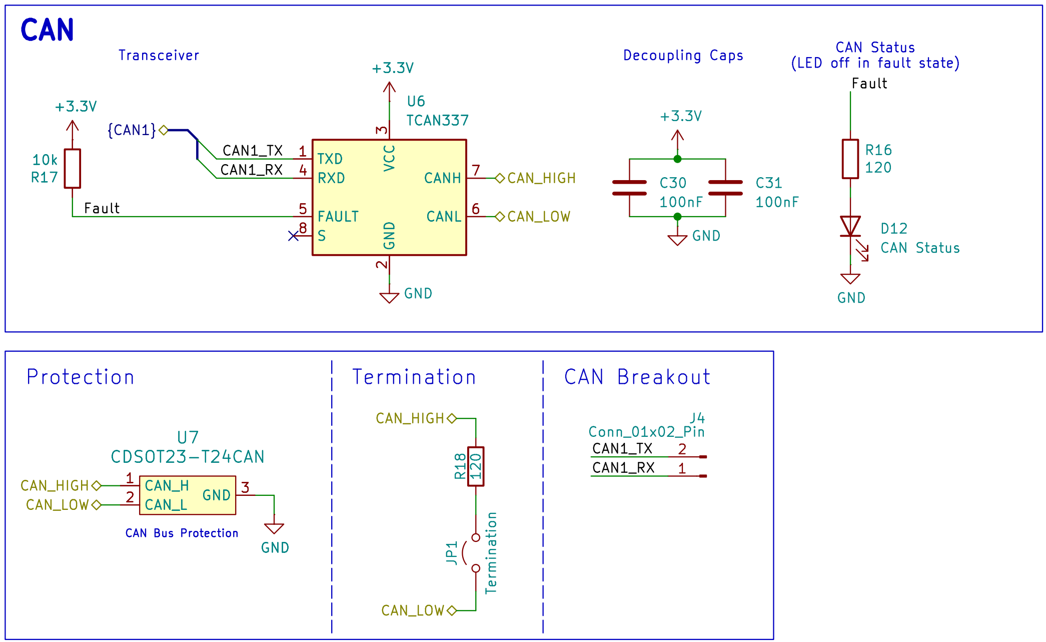

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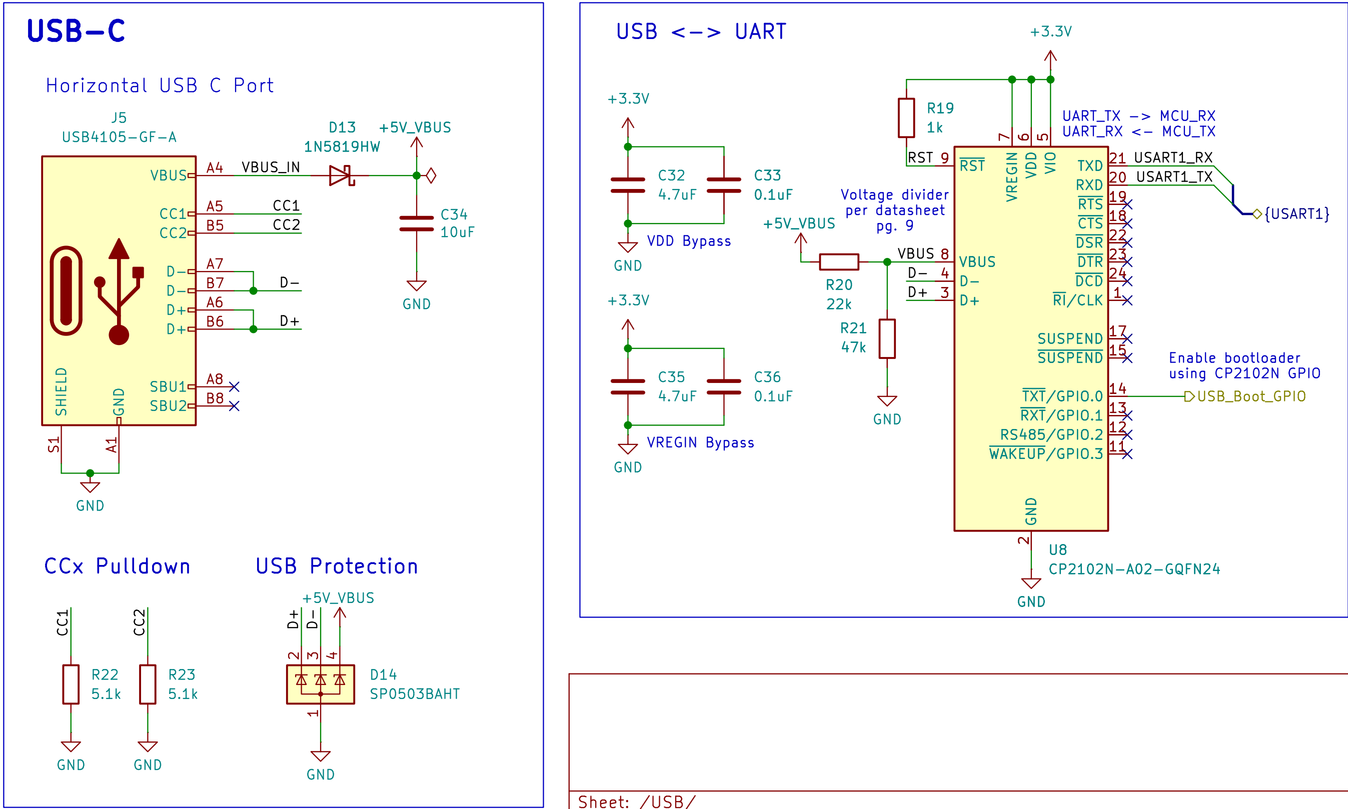

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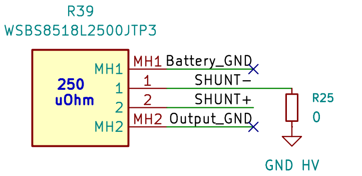
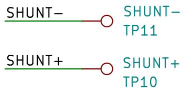
Input

250uOhm Shunt Resistor



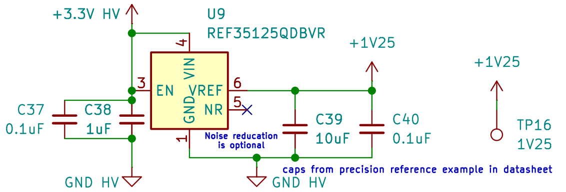


Expected current range: −32 to 70 A
w/ corr. voltage range: −8 to 17.5 mV

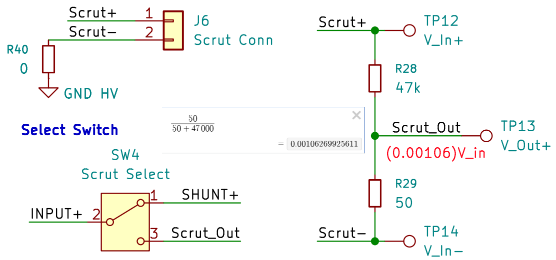
Low−side shunt, so current path is as follows:
Discharge: Output_GND −> Battery_GND
Charge: Battery_GND −> Output_GND

Footprint is M4 version; edit it to have M3 holes

Voltage Reference (1.25V)



Current Scrutineering



DIRECTIONS
− Pass in voltage from scrutineering board
− Use switch to select between shunt / scrutineering inputs

RELATIONSHIP
* I_OUT = ((0.00106)V_in * 100) + 1.25 (V)
* Spoof Current = (0.00106)V_in / (250 uOhm) (A)

Example

Input (V)	Scrut_Out (mV)	Spoof Current (A)	I_OUT Reading (V)
−10 V	−10.6 mV	−42.5 A	0.19 V
+17 V	+18 mV	+72.2 A	3.05 V

1% tolerance on resistors

SHUNT− and Scrut− connected to GND HV to prevent common−mode voltage from being out of INA's range

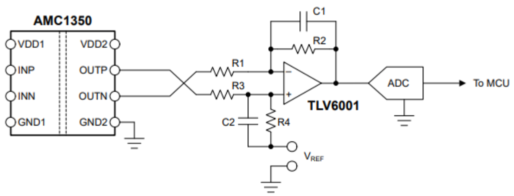


Figure 8-3. Connecting the AMC1350 Output to a Single-Ended Input ADC

With R1 = R3, R2 = R4, C1 = C2:

Gain = R2 / R1

$$g = \frac{24900}{10000}$$

= 2.49

Cutoff Frequency = 1/(2*pi*R2*C)

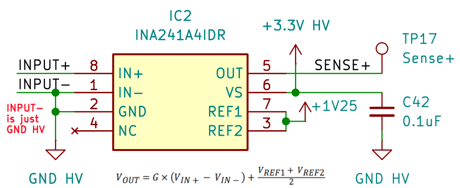
$$f = \frac{1}{2\pi \cdot (24900)\left(68 \cdot 10^{-9}\right)}$$

= 93.9965409236

Current Sense Path

Shunt −> current sense amp −> isolation −> filter

Bidirectional current sense amp



$$V_{OUT} = G \times (V_{IN+} - V_{IN-}) + \frac{V_{REF1} + V_{REF2}}{2}$$

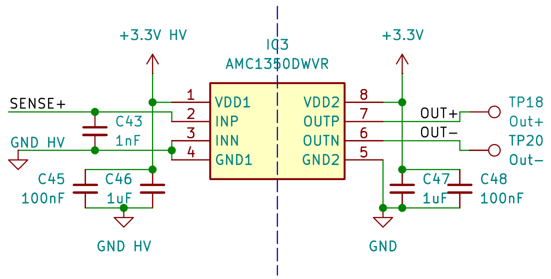
Gain: 100 V/V (INA241 A4 version)

Reference voltage (bias for discharge): 1.25V
1. (70A * 250 uOhm * 100)V + Vref = 3 V
2. (−32A * 250 uOhm * 100)V + Vref = 0.45 V

Maximum current range: −46 to +78A (0.1 to 3.2V output)

Make sure common−mode stuff is within range

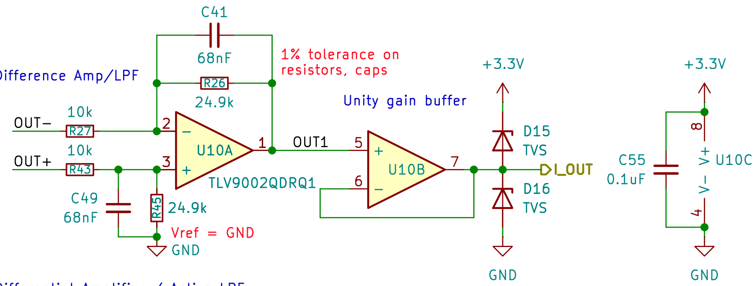
Isolation



Fixed Gain: 0.4 V/V
Target Voltage Ranges:
* Input: 0.1 − 3.2 V
* Output: 0.04 − 1.28 V

OUTN should NOT be tied to GND

Amplification + Low−Pass Filter



Differential Amplifier / Active LPF
* Vout = [(OUT+) − (OUT−)] * Gain
* Gain = 2.49 V/V
* Cutoff freq = ~94 Hz
* 1st order lowpass: −20 db/dec rolloff

Target Voltage Ranges:
* Input: 0.04 − 1.28 V
* Output: 0.01 − 3.19 V

See LPF documentation in Amperes Design folder

Table 8-1. R_{SENSE} Selection and Power Dissipation (1)

PARAMETER		EQUATION	RESULTS AT V_S = 5V				
			A1, B1 DEVICES	A2, B2 DEVICES	A3, B3 DEVICES	A4, B4 DEVICES	A5, B5 DEVICES
G	Gain		10V/V	20V/V	50V/V	100V/V	200V/V
V_{SENSE}	Ideal differential input voltage	$V_{SENSE} = V_{OUT} / G$	500mV	250mV	100mV	50mV	25mV
R_{SENSE}	Current sense resistor value	$R_{SENSE} = V_{SENSE} / I_{MAX}$	50mΩ	25mΩ	10mΩ	5mΩ	2.5mΩ
P_{SENSE}	Current-sense resistor power dissipation	$R_{SENSE} \times I_{MAX}2$	5W	2.5W	1W	0.5W	0.25W

Gain: 100 V/V
Rsense = 250 uOhm
Psense = Rsense * (70A)^2 = 1.225 W

Picked shunt first, then adjusted gain to meet ADC range
If additional gain is needed, tweak filter circuit

Current to I_OUT Conversion

I_OUT = (current * shunt * gain) + Vref (V)

ADC to Current Conversion

Current = ((ADC * (3.3 / 4096)) − Vref) / (Shunt*gain) (A)

Vref = 1.25 V | Shunt = 250 uOhm | Gain = ~100 V/V (100 * 0.4 * 2.49)
Shunt * gain = 250 uOhm * 100 V/V = 0.025

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